



EXTREMES

Most extremophiles are archaeobacteria. More complex organisms may survive in hostile conditions, but only for short periods of time. What they all have in common is that they have developed mechanisms that allow them to handle the extreme environments.

A DOSE OF SALTS



BRINE SHRIMP

Inland lakes that can't drain into the sea slowly build up very high levels of mineral salts in their waters.

These salts can be acidic or alkaline substances, or simply common salt, sodium chloride. Many land organisms can't cope with salty surroundings because it draws water out of their cells and they die. However, some bacteria and algae can live in the salt, along with brine shrimp and brine flies, which are food for birds.

DRY AS DUST



LICHEN

No organism can survive without water, but some can manage with a very small amount. Fungi are the most successful at getting by on very little water, using their root-like hyphae to share moisture and nutrients. Molds are also able to spread quickly across dry foods, such as grains and nuts. Some lichens, too, can live on baking hot rocks in the desert, almost shutting down their growth until it rains. Their spores have thick coats to prevent them drying out.

REAL EXTREMISTS



COFFINFISH

Inhabitants of other extremes include **barophiles**, such as coffinfish, which live in high-pressure conditions, mainly at the bottom of the sea. **Radioresistants** can cope with levels of radiation that would kill other organisms. **Anaerobic extremophiles**, such as the bacteria that live in the guts of animals, are able to live without oxygen. **Polyextremophiles** can even survive several different extremes at once.